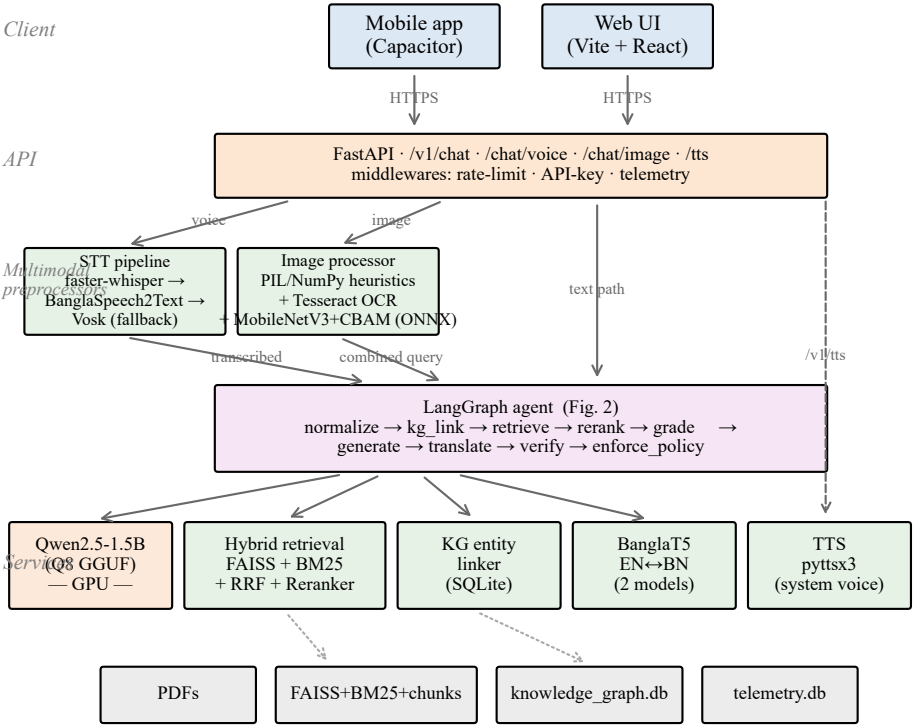

Client
Mobile app
(Capacitor)
Web UI
(Vite + React)
HTTPS
HTTPS
API
FastAPI · /v1/chat · /chat/voice · /chat/image · /tts
middlewares: rate-limit · API-key · telemetry
voice
image
Multimodal
preprocessors
STT pipeline
faster-whisper →
BanglaSpeech2Text →
Vosk (fallback)
Image processor
PIL/NumPy heuristics
+ Tesseract OCR
+ MobileNetV3+CBAM (ONNX)
text path
transcribed
combined query
/v1/tts
LangGraph agent (Fig. 2)
normalize → kg_link → retrieve → rerank → grade →
generate → translate → verify → enforce_policy
Services
Qwen2.5-1.5B
(Q8 GGUF)
— GPU —
Hybrid retrieval
FAISS + BM25
+ RRF + Reranker
KG entity
linker
(SQLite)
BanglaT5
EN↔BN
(2 models)
TTS
pyttsx3
(system voice)
PDFs
FAISS+BM25+chunks
knowledge_graph.db
telemetry.db